Deep learning to identify ECG features associated with risk of ischemic stroke and atrial fibrillation

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Primary questions

Can ECG data improve stroke predictions based solely on clinical covariates?

How much is gained by incorporating deep learning compared to simpler methods (logistic regression)?

And can we identify novel ECG features that add to our understanding of ischemic stroke?

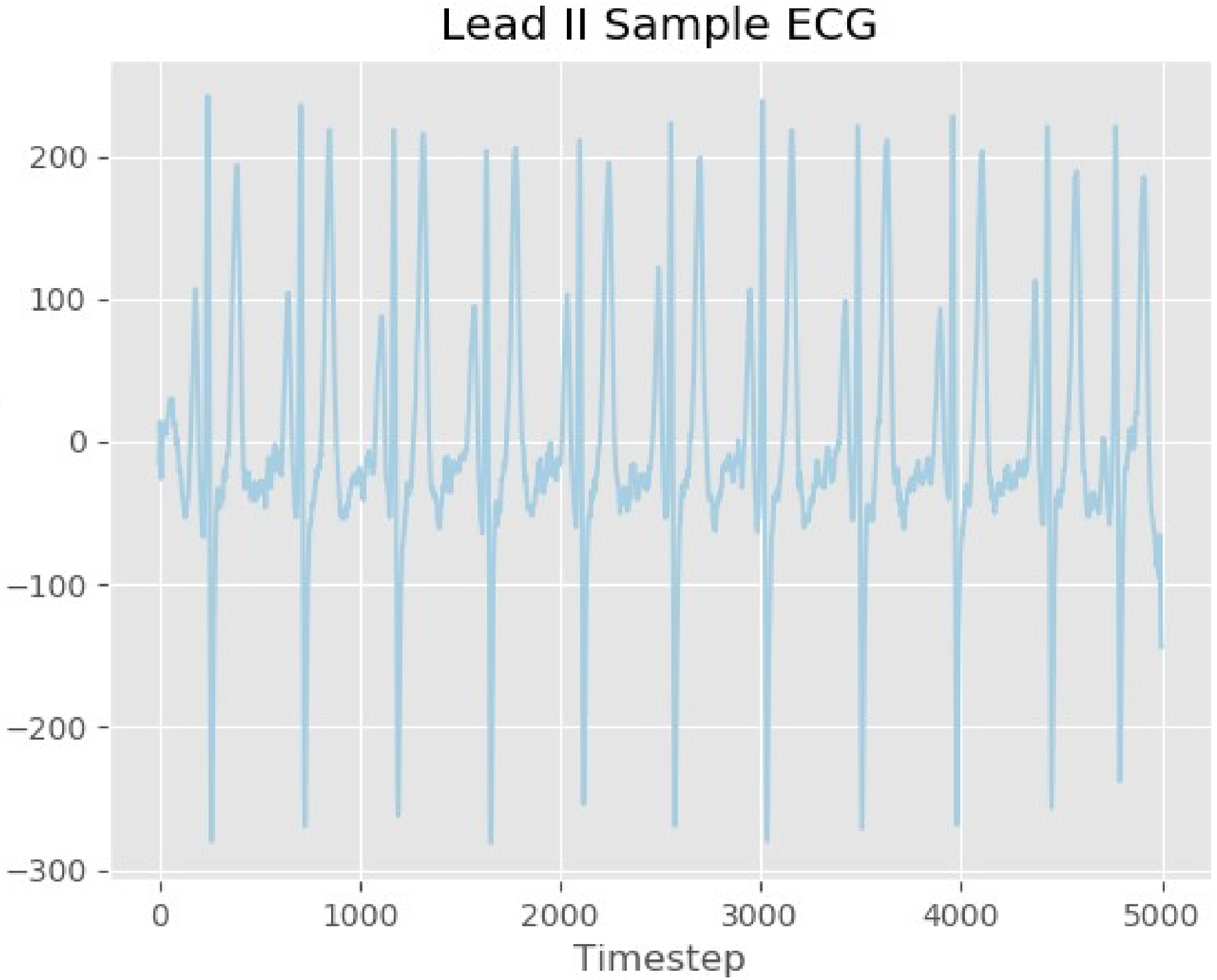
Patient data

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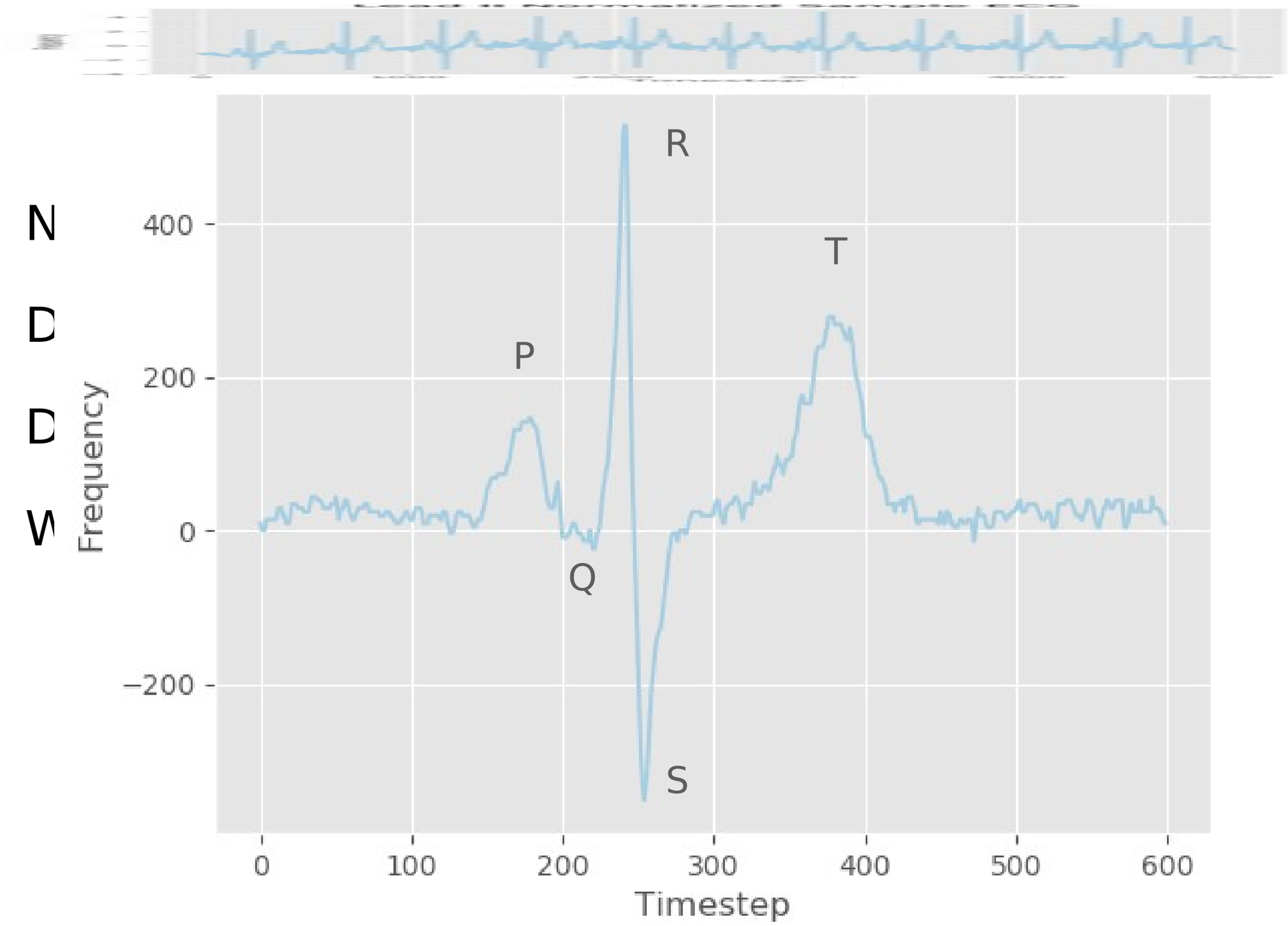
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Patient data



Patient data

Exclusions

Data from the Cardiovascular Health Study (CHS) contains serial 10-second, 12-lead ECG data and Marquette readings

Total patients in dataset (5888)

- Patients with record of prior stroke at baseline (249)
- Patients with a pacemaker at baseline (3)
- Patients with Atrial Fib on baseline ECG (63)
- Patients with no baseline ECG (210)
- ECGs with EPICARE baseline score of 5 (334)

5029 patients eligible for the study

Patient data

Clinical covariates

Age (at baseline)

Systolic blood pressure (mmHg)

Adjusted cholesterol (mg/dl)

Gender

Diabetes - defined as fasting glucose>=126, non-fasting glucose>=200, or use of a glucose-lowering medication

Prior CHD (at baseline)

Use of any hypertension medication

Prior heart failure (at baseline)

Body mass index

Smoking status

P-wave terminal force in V1 (PTFV1)

Patient data

Missing data

After Exclusions	5029
BMI	15
Systolic blood pressure (mmHg)	6
Hypertension medication	6
Diabetes: elevated fast/non-fast glucose or use of DM meds	65
Adjusted cholesterol (mg/dl)	44
Smoking status	6
Total	4924

Patient data

Summary statistics of clinical covariates

		Mean	Median	Std	Range
Age		72.6	71	5.5	65-100
PTFV1		2693.8	2577	2196.8	0-19688
Cholesterol (adjusted)		211.7	209.8	39	58.8-429.8
Systolic blood pressure		136.2	134	21.6	79-234.8
BMI		26.6	26.1	4.7	14.7-58.8

Patient data

Summary statistics of clinical covariates

		Proportion
Gender		42% male
Diabetes: elevated fast/non-fast glucose or use of DM meds		14.000000%
Prior heart failure at baseline		3.500000%
Prior CHD at baseline		18.000000%
Smoking status		46% non, 42% former, 12% current

Patient data

Stroke incident rates

Of the 4924 patients included in the dataset, a total of

738 had strokes (15.0%)

431 within 10 years (8.8%)

213 within 5 years (4.3%)

41 within 1 year (0.8%)

Why CNNs?

If we knew which parts of the ECG or what features indicate stroke, our task would be simple - engineer those features and input them into a generalized linear model

Need to find the “important” parts of the ECG and identify the key features for the prediction task

Relative location is important - the data is ordered/linear

There is temporal information we need to capture, and multiple time series per person - how those leads engage together

Scale or absolute location may vary (shift-invariant)

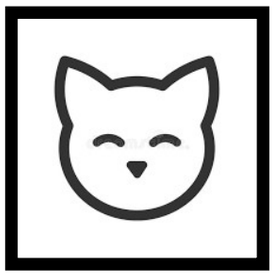
Stroke is a complex pattern to identify, want to build combinations of smaller and simpler patterns to capture it

By stacking convolutional layers that learn meaningful local filters, can capture large complex patterns (stroke)

Why CNNs?

Image Classification

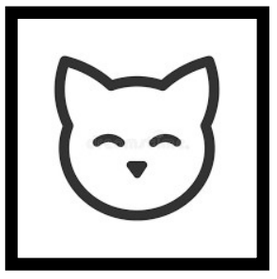




Why CNNs?

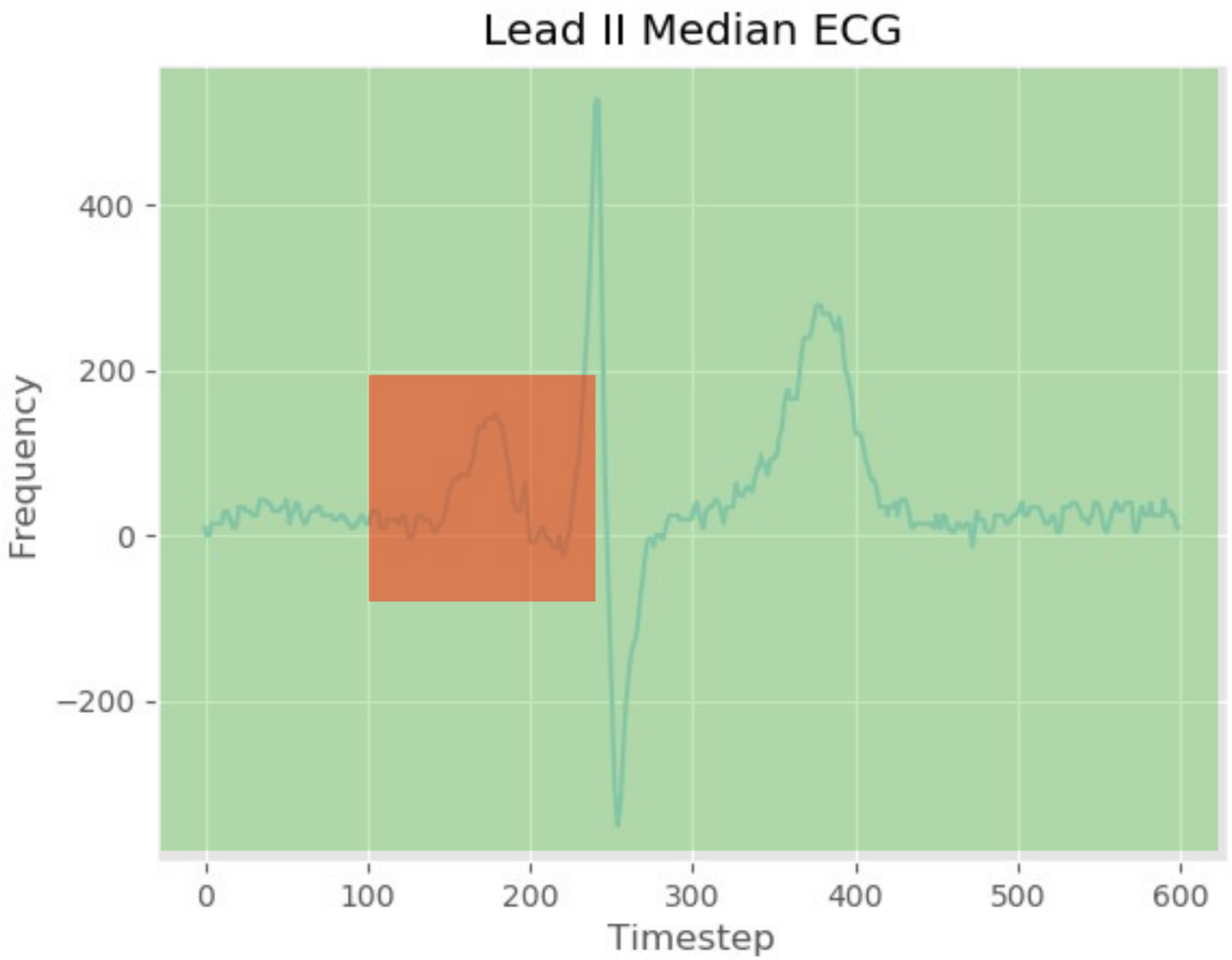
Image Classification





Why CNNs?

Image Classification



Extended P-wave

Analysis

Basic setup

Data was split into 80% training and 20% testing, performed 5-fold cross validation on training data with stratified sampling by class

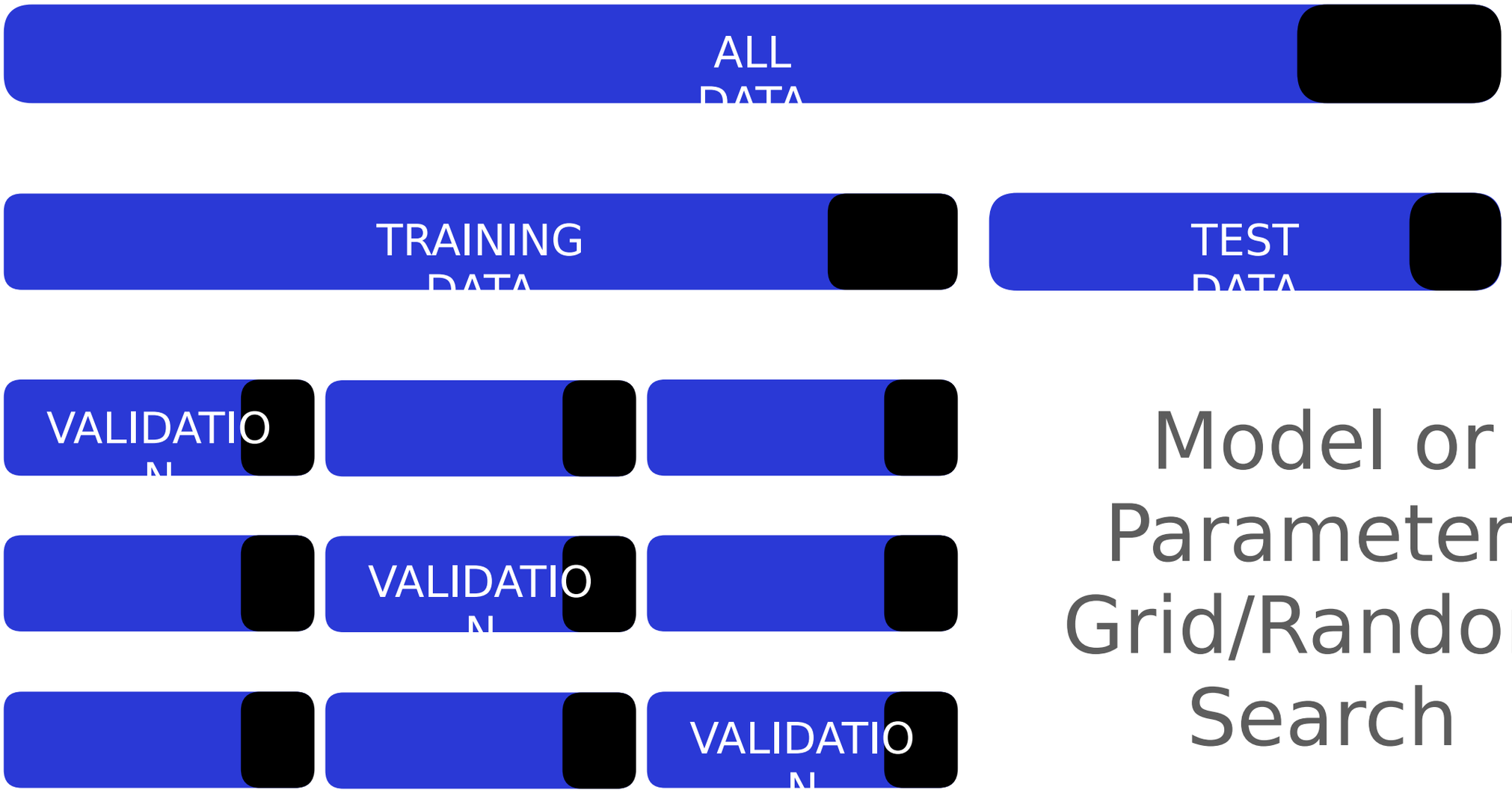
Considered a number of factors in terms of model performance, but focused on AUC

Aimed at predicting risk of stroke within 5 years from baseline ECG and common clinical covariates, but looked at different prediction windows

Began with standard CNN architectures and parameters, then experimented with deeper networks and recurrent layers

Analysis

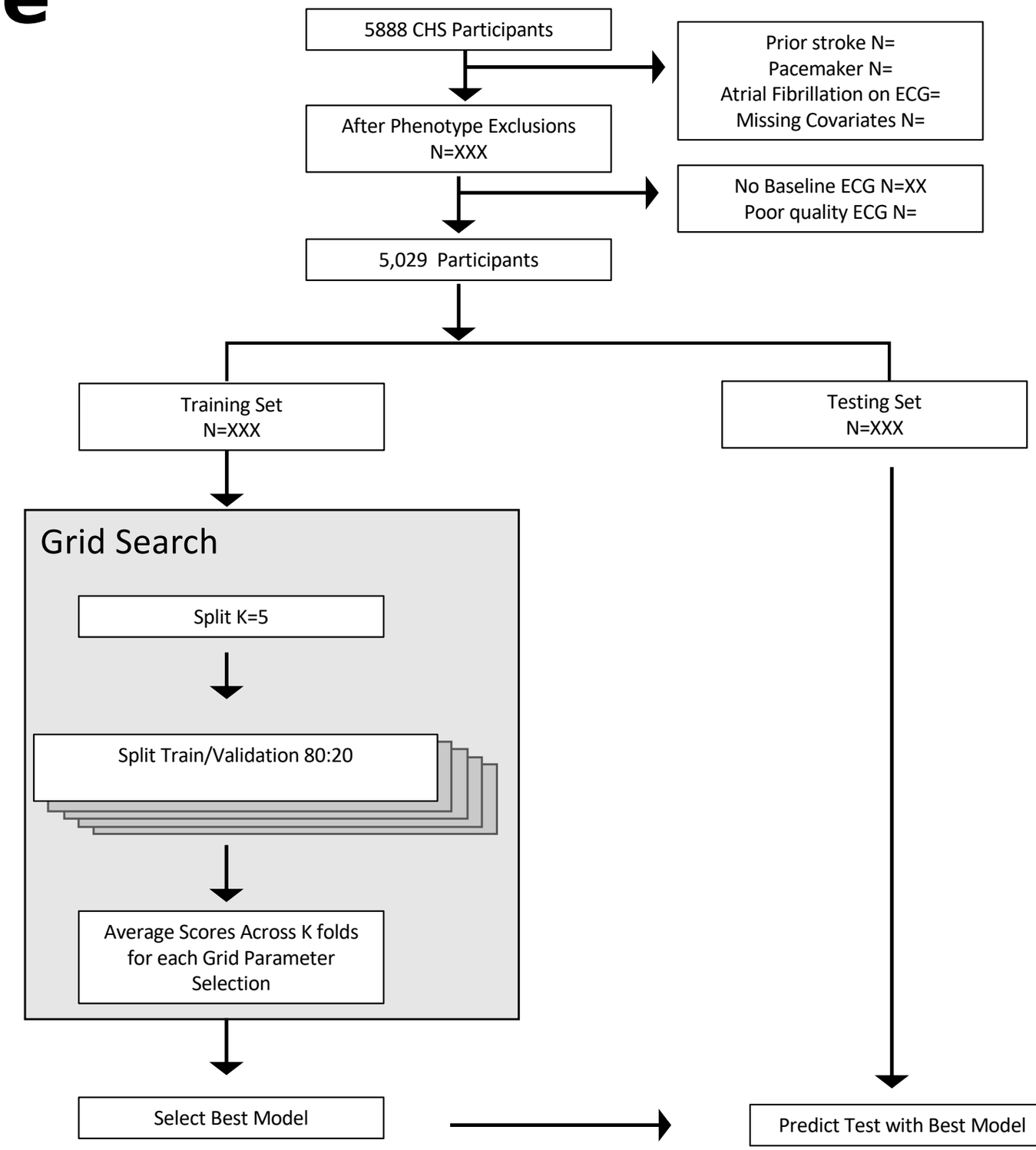
Stratified K-Fold Cross Validation



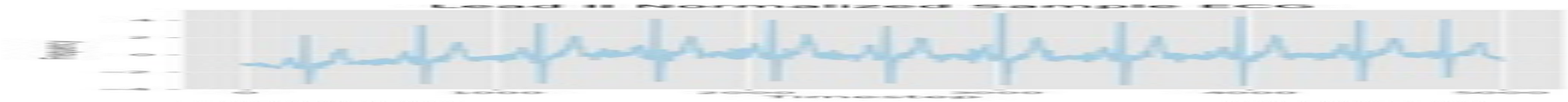
Model or Parameter Grid/Random Search

Analysis

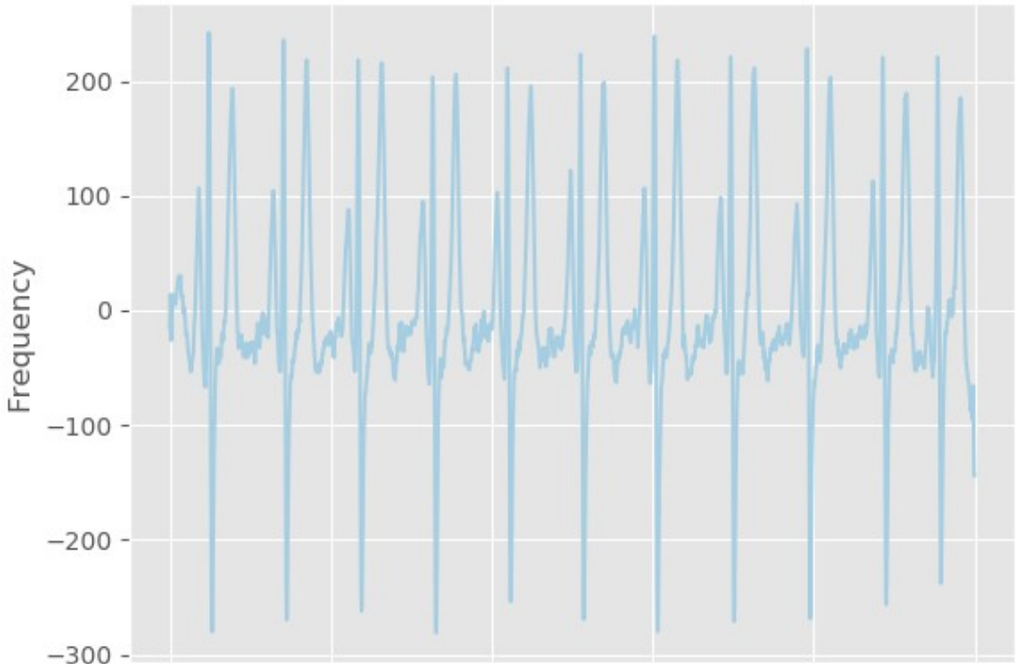
Pipeline



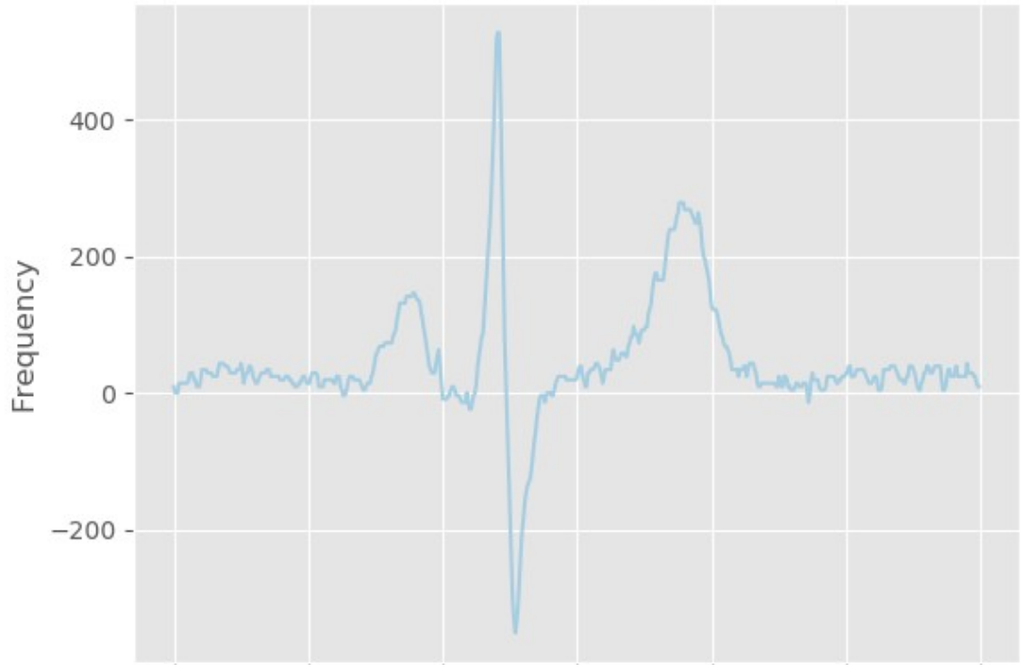
Analysis



Lead II Sample ECG



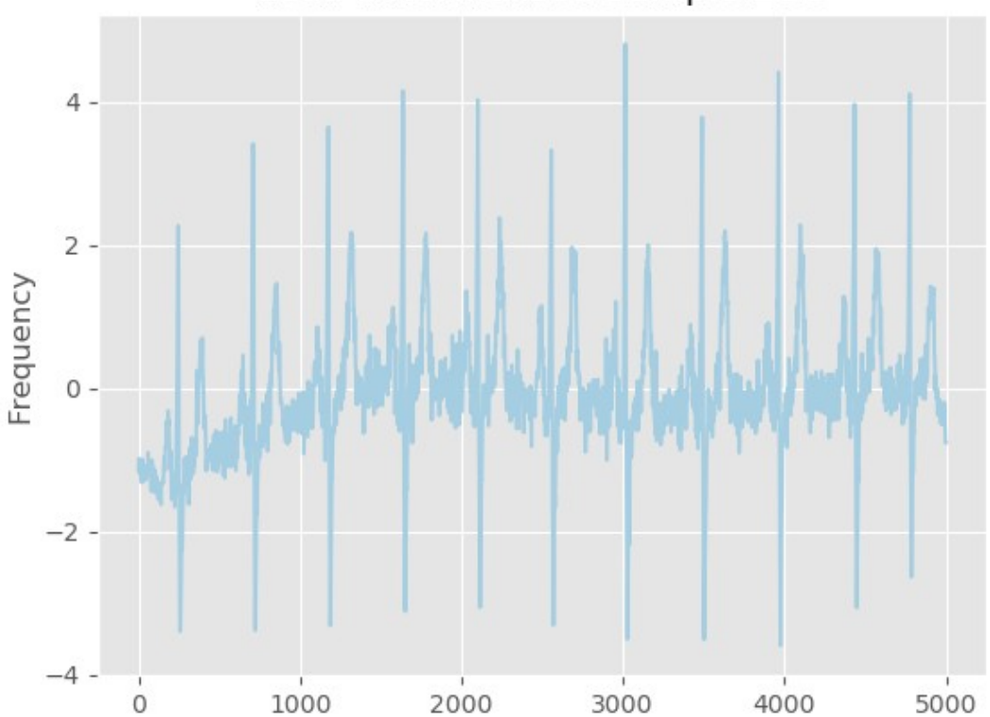
Lead II Median ECG



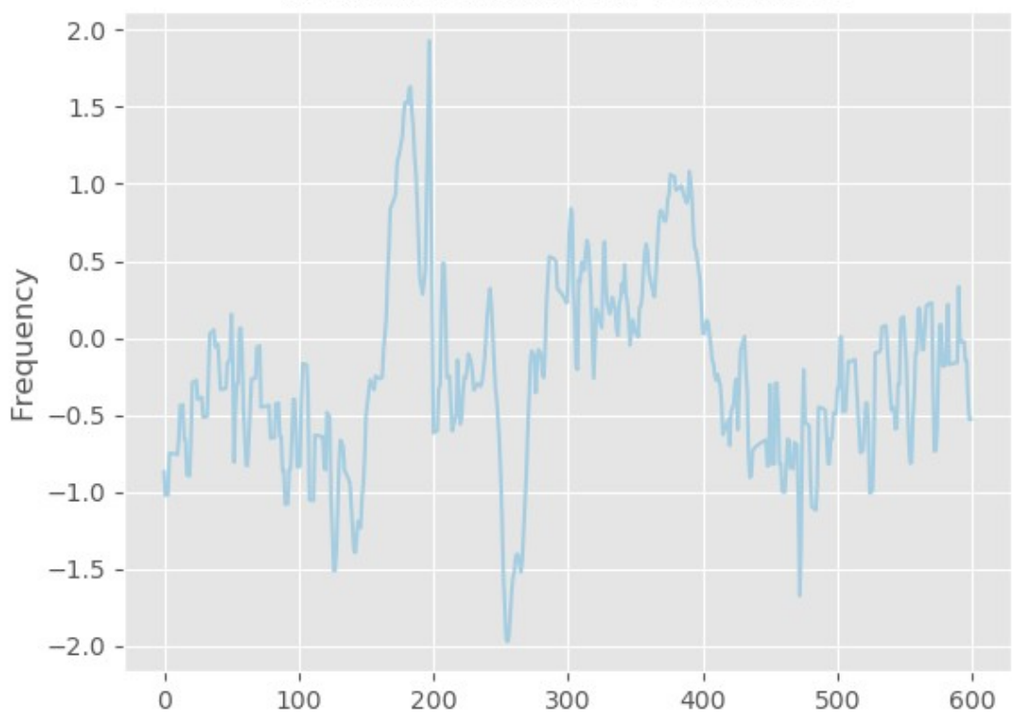
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Lead II Normalized Sample ECG

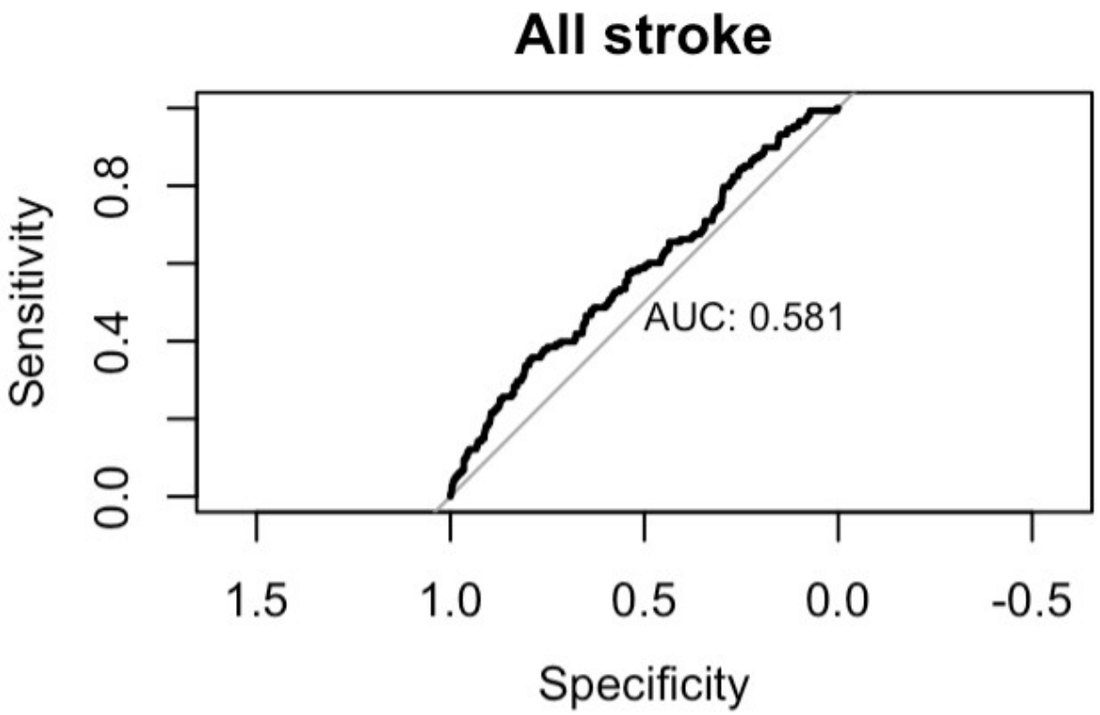


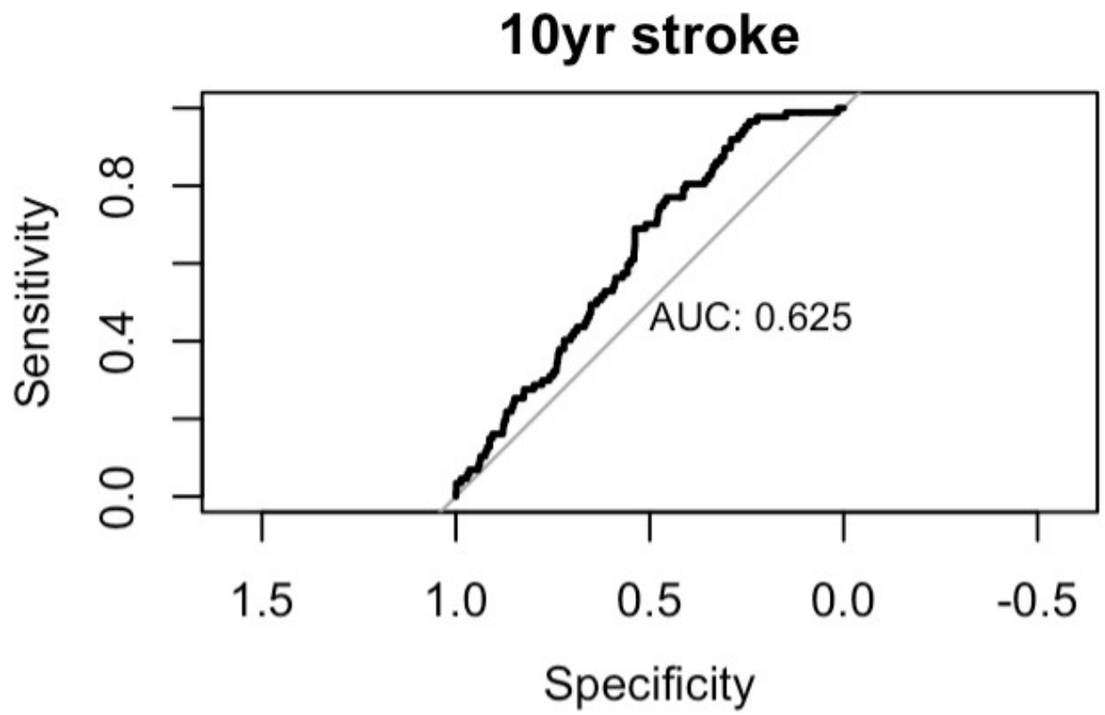
Lead II Normalized Median ECG

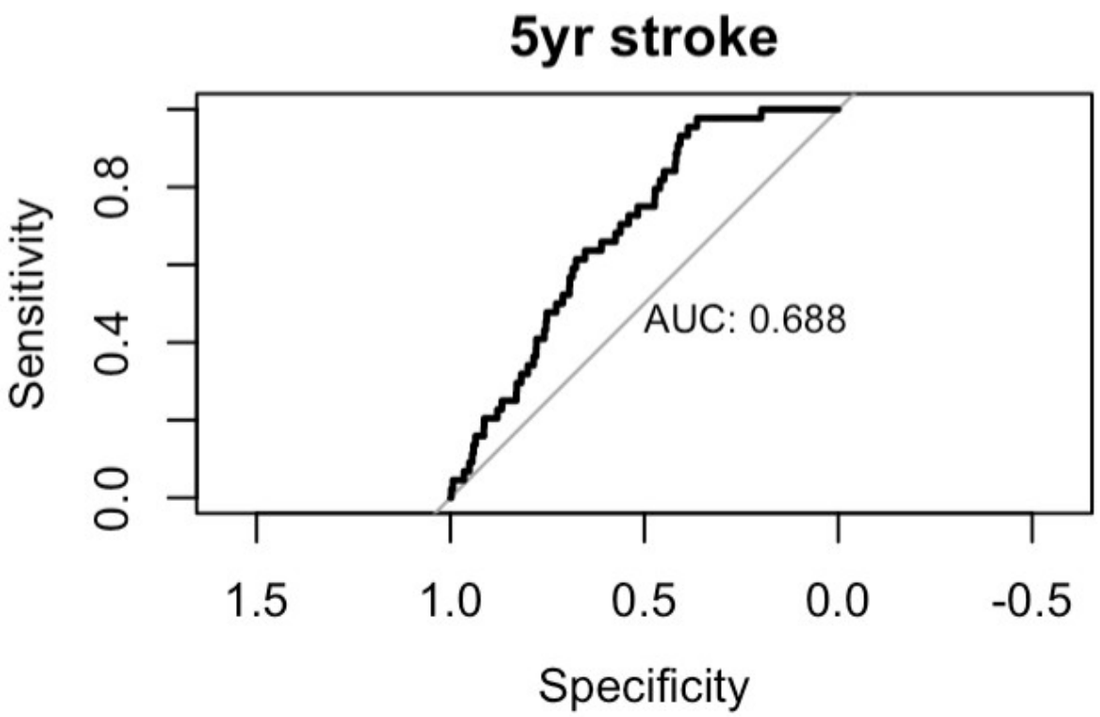


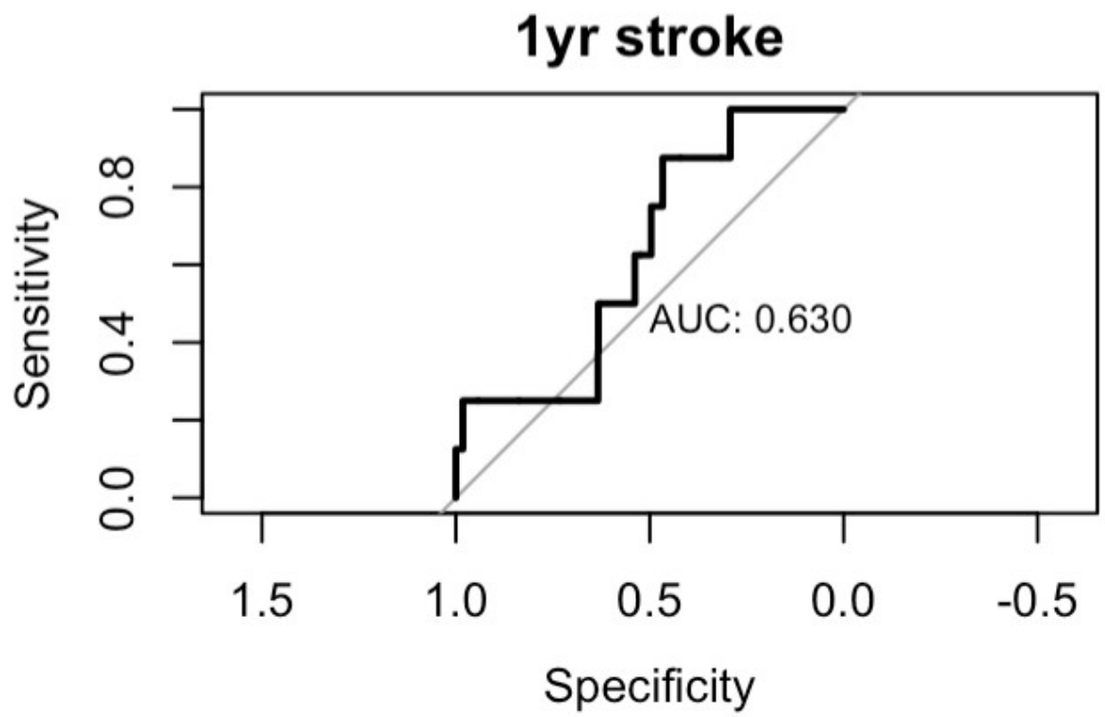
Results

Logistic regression on clinical covariates test AUC









Results

Test AUC comparison for 5yr stroke

Input	AUC
Median	0.65 - 0.68
Normalized Median	0.68 - 0.73
Rhythm	0.62
Normalized Rhythm	0.63 - 0.64

Results

Comparing number of leads, ECG only

	Median		
	1 Lead (Lead II)	**8 Leads**	**12 Leads**
All	0.57	0.56 - 0.58	0.57
10 yr	0.59	0.61	0.61
5yr	0.58 - 0.62	0.65 - 0.68	0.65 - 0.68

Results

Comparing number of leads, ECG plus covariates

	Median		
	1 Lead (Lead II)	**8 Leads**	**12 Leads**
All	0.57 - 0.58	0.59 - 0.60	0.60 - 0.61
10 yr	0.6	0.65 - 0.66	0.65
5yr	0.67 - 0.70	0.73 - 0.74	0.73 - 0.74

Results

Test AUC comparison across timeframes

Timeframe	ECG	Covariates	ECG + Covariates	Logistic Regression
All	0.59	0.6	0.60 - 0.61	0.58
10 yr	0.62	0.64 - 0.65	0.65	0.63
5yr	0.68 - 0.73	0.71 - 0.72	0.76 - 0.77	0.69

Key takeaways

Clinical covariates outperform ECG data alone, but the best performance uses both as input

Normalizing ECGs increased performance across all models, and outperformed more sophisticated preprocessing methods

Median heartbeat has performed better than any version of the rhythm ECG data, and requires lower computation time

Best model results have been from simple CNN architectures and outperformed RNNs and CRNNs

Challenges

Reproducing AUCs reported in previous papers - additional preprocessing, incomparable datasets, different prediction criteria

Prediction timeline - determining appropriate prediction times and long-term prediction limits

Custom data generators to handle the size of all ECG files, different formats, and various model architectures